SIMBLOCKS // RESULT

BATTERY FOR A 180 S FIGHT

What is the smallest 6S pack that finishes a full fight?

2026-10-16 08:03Z

● PASS VERIFIED

MINIMUM 6S PACK (95% OF FIGHTS) PASS

1500 mAh

at 3.3 V/cell; median fight 1196 mAh

CHARGE USED PASS

SF 1.87

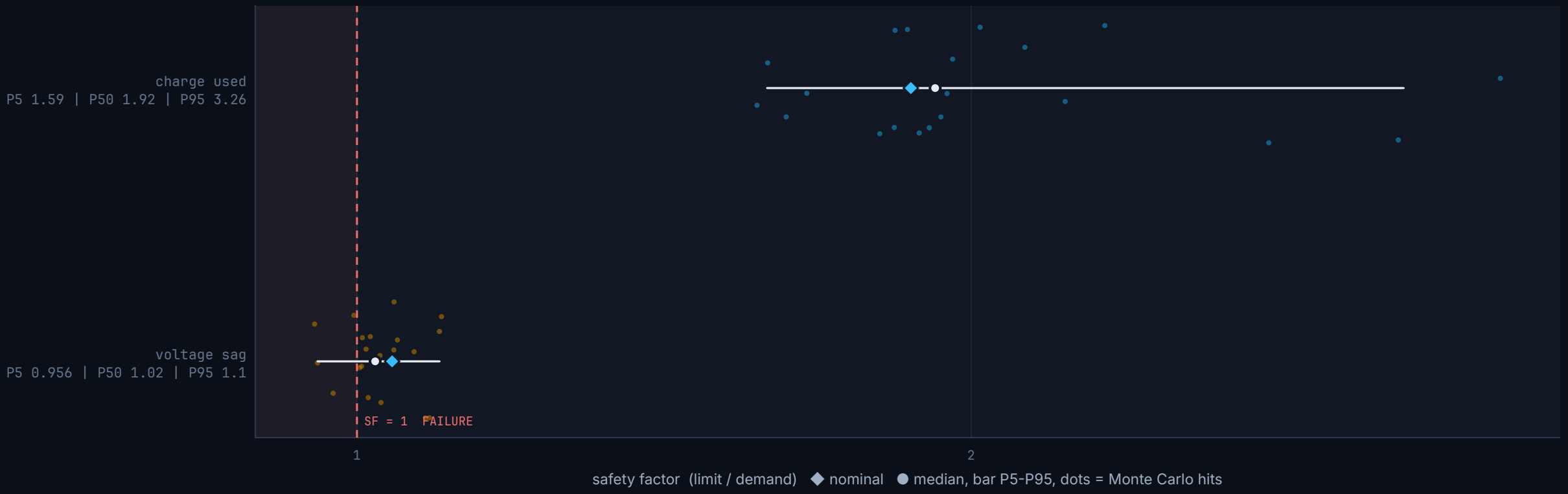
P(fail) 0% | SF P5 1.59

VOLTAGE SAG [GOVERNING] WARN

SF 1.04

P(fail) 20% | SF P5 0.956

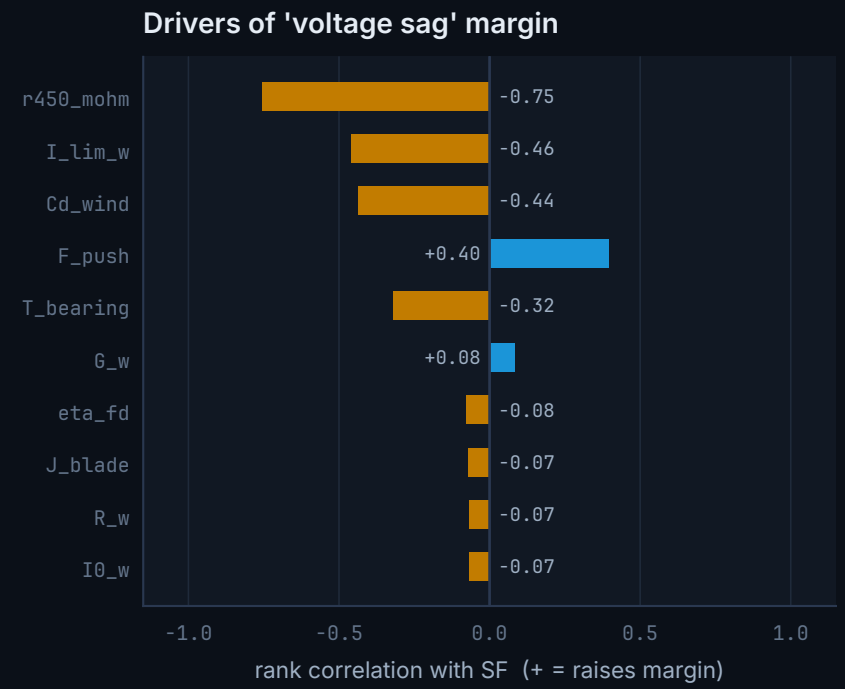
Margin to each user-defined failure mode



Battery over a 180 s fight: VERIFIED | PASS 14 WARN 2 FAIL 0 SKIP 0 | 51 runs, 7.0 s

BATTERY FOR A 180 S FIGHT

What to design around: drivers of the governing margin and the design-variable sweep



SIMBLOCKS // VERIFICATION

2026-10-16 08:03Z

BATTERY FOR A 180 S FIGHT

● PASS VERIFIED

Automated verification battery

CHECK	STATUS	DETAIL
determinism	● PASS	bit-identical repeat
finite_outputs	● PASS	all outputs finite
physical_bounds	● PASS	3 bounds respected
stability_margin	● PASS	critical dt 0.000867 s / dt 0.0001 s = 8.7x
dt_convergence	● WARN	Wh: err 9.1e-05% (order -0.53); mAh_used: err 9.8e-05% (order -0.55); v_min_cell: err 0.00091% (order 0.7); I_peak: err 0.0059% (order 0.96); t_fir...
energy_ledger	● PASS	worst closure residual 0.41% across 3 dt levels (tol 5%)
limit:idle only: 3 W x 180 s = 0.15 Wh	● PASS	matches expected limit
knockout:no blade windage	● PASS	Wh -26.8%, mAh_used -27.7%, v_min_cell +2.04%, I_peak -0.978%, t_first_full -3.48%
knockout:no weapon hold (governor off)	● PASS	Wh -41.5%, mAh_used -42.6%, v_min_cell +4.03%, I_peak -1.46%, t_first_full +0%
knockout:no pushing	● PASS	Wh -26.5%, mAh_used -27.4%, v_min_cell +3.41%, I_peak -4.02%, t_first_full +0%
knockout:no normal driving	● PASS	Wh -8.82%, mAh_used -9.17%, v_min_cell +3.8%, I_peak -14.3%, t_first_full +0%
knockout:no hits (one spin-up only)	● PASS	Wh -13.3%, mAh_used -13.8%, v_min_cell +4.69%, I_peak -27.6%, t_first_full +0%
knockout:no pack recovery circuit (R1 = 0)	● PASS	Wh -0.52%, mAh_used -0.542%, v_min_cell +1.25%, I_peak -1.24%, t_first_full +0%
sensitivity	● PASS	10 inputs swept low/high (trends reported, not judged)
monte_carlo	● PASS	20 samples, 0 failed finite/bounds/ledger (0.0%)
failure_modes	● WARN	charge used: nominal SF 1.87, P(fail) 0%; voltage sag: nominal SF 1.04, P(fail) 20%